

Professional Synopsis

Around 12 years of experience in facilitating analytical engagements in field of Data Solutioning stack and worked on live projects with clients in various domains such as Healthcare, Insurance, Telecom, Cyber security, Industrial, Automotive, Educational Institution etc. across multiple geographies in US, Europe and Asia

Certifications:

- ✓ AWS Certified Solutions Architect Associate - SAA C03
- ✓ Certified Big Data Engineering and Machine Learning Specialization from IIT Roorkee

Solved various industrial problems and provided consulting solutions in domain of Data such as data engineering, governance and compliance, analytics and visualization, storage and management, data science and machine learning for companies like BMW, Roche-Genentech, Reliance, Aria-Jefferson Health, Maybank, Indigo, Prudential, Fidelity International and numerous other global industry leaders. Leveraged innovative approaches and cutting-edge technologies to address complex issues, including but not limited to process mining, market basket analytics, data lake development, ETL development, fraud identification with outlier modelling, recommendation engines, Lidar Data Analytics, IOT, Image modeling, Vision Processing, GPS, LSR, probes userplane data modeling, among others

Extensive experience in Big Data Technology, data processing pipelines and data warehouse aspects using tools such as Hadoop (Cloudera, MAPR, Horton Works), AWS Glue, AWS EMR, Athena, S3, AWS Lambda, GIT Lab and Bitbucket, GCP cloud Dataflow, CI/CD, Agile, Scrum -Jira

Leveraged expertise in Azure services, including ADF, Databricks, Azure DevOps, ADLS, Power BI, HDInsight, ADLS, Event Hub, Logic App, and Synapse, and contributed to Agile Framework projects

Led the design and implementation of scalable, secure, and cost-effective data lake/delta lake/data warehouse solutions on Azure Cloud, including the migration of on-premises EDW to Azure Cloud. Created Proof of Value (PoV) architectures and provided strategic recommendations to clients.

Specializes in automating various capabilities within the Azure Synapse platform, specifically focusing on SQL Server database development tasks such as creating tables, views, stored procedures, and functions.

Involved in data governance, data security and enrichment techniques like data domain discovery, business glossary, data profiling, data quality etc.

Demonstrates proficiency in conducting rigorous unit testing and providing comprehensive support for Quality Assurance and User Acceptance Testing (UAT) processes.

Hands-on experience in using tools and techniques such as Pyspark Ingestion and Modelling Jobs including Ensemble modelling, Decision Trees, Regression modelling (Linear & Logistic Regression), SARIMA (Time series modelling), clustering using K means etc. Performed descriptive, predictive and prescriptive analytics and developed Dashboards representing measurable KPIs on Tableau and Sagemaker with Activities: Data extraction, cleaning, model building, validation, researching and redesigning

Programming Skills/Software used: Python (Pycharm, Jupyter notebooks), SSIS, VBA, Node.JS, Docker, Container, Kubernetes, Scala, R Studio, SAS, Knime, Cloud (AWS, GCP, Azure), Airflow, Socket.IO, MDM, PIM, Informatica PowerCentre, EDC among others

Experience in handling Databases SQL & No SQL databases such as MSSQL, My SQL, HDFS, AWS Redshift and Dynamo DB

Strong Problem Solving, Client and Project Management skills with experience in leading and working with cross functional/cross location teams of 100+ Associates working on multiple analytics projects

WORK HISTORY AND EXPERIENCE

Freelance

Present

A freelancer specializing in optimizing data and AI strategies, offering comprehensive end-to-end platform governance, security, and development services.

StatusNeo Technology

Principal Consultant (Jun-22 –Aug 24)

As a Principal Consultant worked on various data initiatives involving proposal development, technical solution development, resource planning and allocation, technical deliveries including POC development and delivering BAU activities. Involves training of resources to meet project deadlines. Following projects delivered:

Telecom Client

1. Enhance telecom user experience by creating a next-gen solution that predicts user location and provides intelligence for automated grid allocation. A machine learning model along with completely automated process was developed with optimized data operations, and architecture to make location predictions more efficient and cost effective resulting in > 90% savings in resources by reducing manual intervention.
2. Spearheaded the development and implementation of a machine learning algorithm for LSR and probes userplane data with the intent of developing hyperlocalization solution.

Vision Team

1. Developed a combined model for 3D shape completion by integrating a pretrained classifier with a convolutional neural network (CNN) architecture.
2. Efficiently preprocess and manipulate volumetric data represented as voxels using techniques like clipping, masking, and reshaping.
3. Model Architecture: The combined model consists of a pretrained classifier to extract features and a CNN for learning complex patterns, enabling accurate shape completion in voxelized data.
4. Training and Evaluation: Train the model using voxelized shape datasets, optimizing a custom loss function, and evaluate performance using metrics like mean squared error (MSE) and mean absolute error (MAE).

Advertisement Team (POC Phase)

1. Real-time Advertisement Processing: Utilizes AdFetchProcessor, MallVicinityProcessor, and ScoreAdsProcessor to manage real-time ad data, covering tasks such as fetching ads, identifying nearby malls, and recommending ads to users.
2. Integration and Interaction: Employs KafkaConsumer and KafkaProducer for message handling within processors, alongside interactions with external APIs for fetching ad data based on user and mall information, ensuring dynamic ad retrieval and scoring.
3. Data Processing and Scalability: Processes incoming data from Kafka topics, transforming it within each processor to generate tailored ad recommendations based on user preferences, while featuring a scalable and modular architecture for easy addition or modification of processors to meet future requirements.

Other Clients

1. Spearheaded the blueprinting, creation, and implementation of data confluence and ETL solutions within a data warehousing and BI environment
2. Exhibited adeptness in JDBC elements, context genesis, dynamic assignment of contexts, and assorted functions for date and string manipulation.
3. Applied profound comprehension of database principles, including data pattern, standardization, dimensionality, referential integrity, indices, keys, master data, and metadata.
4. Demonstrated strategic insight, contemplating the broader business repercussions of technical determinations
5. Advocated for simplicity in solutions and blueprints, favoring straightforward strategies over intricacy.
6. Proactively identified enhancement opportunities and embraced change when requisite.
7. Maintained equanimity and orderliness in fast-paced and demanding scenarios.

Saama Technologies

Technical Lead (Jul-20 –Jun-22)

1. Big data analytics lead responsible for setting up end to end big data analytics and science capabilities and solutions for the pharma clients. This includes data ingestion pipelines for big data using pyspark on aws services like EMR and Glue.
2. Storing the data in compressed formats like parquet and ORCs, data partitioning, S3 bucketing strategies, KPI development, visualisation on Tableau and then doing data science (majorly forecasting) using Jupyter notebooks.
3. Automating the data ingestion and analytics pipelines using Apache Airflow DAGs and scheduling Tableau reports.
4. Also Responsible for mentoring and hiring technical folks to deliver the big data solutions for the company.
5. Evaluated accomplishments in terms of business efficacy, aligning technical endeavors with corporate objectives.
6. Nurtured a DevOps ethos, assuming accountability for comprehensive system evolution and operation.
7. Collaborated cross-functionally, engendering trust and respect across all tiers of colleagues.
8. Prioritized mentorship and actively contributed to the professional development of team members.

YOLOBUS

Data Scientist (Apr-20 - Jun-20)

Lead data scientist responsible for setting up data science capabilities and solutions for the company. Responsible for mentoring and hiring data analyst and associate data scientist to deliver the data science solutions for the company. Identifying the data science use cases that will add value to the company's growth and deliver the solutions.

HCL Technologies

Technical Specialist (Feb-18- Apr-20)

Worked as machine learning consultant and presales specialist, responsible for front-ending technical discussions and presentations with B2B clients including requirement gathering, understanding the current IT setup, Business case, RFPs, Solution Architecture for machine learning, IoT and AI projects. Schengen Business Visa Holder. Also part of cybersecurity team to classify attacks like SQL Injection, DoS, XSS, Malware, Phishing etc.

Projects Undertaken

Education Analytics- Enrolment Forecasting (End to End development)

- ✓ Client requirement: To analyse number of students that may enroll for a particular course term
- ✓ Build Propensity based model to predict the propensity to join a particular program
- ✓ Data from multiple sources: LMS (Learning management System), Powerfaids (Financial Aid etc.)
- ✓ Complete end to end development executed hands-on from Data Infrastructure setup, ETL Pipelines development (spark jobs), Data cleaning and feature engineering, Propensity models development (Survival Model), model validation and Model Deployment as a Rest API in Google App Engine
- ✓ Technology Used: **Google Cloud Platform, Python**

Autonomous Driving Project

- ✓ ADAS Business Requirement: To analyse ethernet signals produced in ADAS (advanced driver assistance systems) and predict signal errors
- ✓ Data was given in MDF4 format which is binary and hierarchal in nature with channel and data groups
- ✓ Data parser was written which extracted the signals and time data in python
- ✓ Data was then stored in MAPR cluster
- ✓ Random Forest model was developed for signal error prediction and case study presented with results in capability deck

COGNITIO ANALYTICS LLC

Sr. Analyst (Sep-15-Jan-18)

As one of the initial members of Cognitio, built in-house healthcare analytics team from scratch, provided training, designed various POC as part of business development and managed live projects worth million \$ across engagements.

Projects Undertaken

Analytics for Healthcare giant in Pennsylvania

- ✓ Understanding business requirements, gathering and assembling required information and performing comprehensive performance analysis for US based Healthcare client and providing a platform for strategic decision making using Analytics and Business Intelligence (Dashboard Development)
- ✓ Delivered **30+ Real-time custom dashboards** representing performance, operational and financial metrics
- ✓ Manage client communication, expectation and ensure hassle free business deliveries in agreed time frames
- ✓ Develop project scopes and objectives, involving all relevant stakeholders, and ensuring technical feasibility
- ✓ Tools: **Python, Lawson BI**

Budget Forecasting for Operation's Management

- ✓ Objective was to develop efficient budget system for the client from traditional method of using last year's number as placeholders for current budget
- ✓ POC was presented and received signoff from the client on developing budget system for various business metrics
- ✓ Built SARIMA Model to predict the budget for FY-18 budget for various operational (LOS, inpatients, admissions and discharges) and financial metrics (Hospital cash)
- ✓ Business Impact: Lowering **MAPE from ~30% to 5%** insured better resource allocation and management for the Hospital
- ✓ Worked on multiple analytics projects majorly for clients in Telecom and Banking domain.
- ✓ Technologies Used: Used R packages "tseries" and "forecast", ADF test, ACF and PACF Plots, Built in methods Holtwinters and Snaive method
 - o Tools used: **R and Power BI**

ANALYTIXLABS

(Dec-14-Aug-15)

Worked on multiple analytics projects majorly for clients in Telecom and Banking domain.

Projects Undertaken

Customer Clustering for Telecom Client

Developed segmentation model for **Telecom** client using clustering algorithm

- ✓ Using K-Means ++ algorithm formed **usage based cluster** for a telecom giant in India
- ✓ Customers were classified into 4 different clusters based on Total Monthly Calls, Internet Usage, 3-way calling (Conference Service) and Total Revenue customer

Predictive Model Based on Technical and Commercial Parameters

Built the **pricing model** based on various technical and commercial parameters of the car

- ✓ Used Linear regression technique
- ✓ Tools Used: **SAS and Excel**

Predicting borrower's chance of defaulting on credit loans

Using predictive modelling technique identified customer who are at high risk of defaulting

- ✓ Performed feature engineering, Univariate and Bivariate analysis, variable reduction steps
- ✓ Used Gradient Boosting techniques (GradientBoostingRegressor) for regression analysis
- ✓ Built Logistic model on SAS

JK TYRES AND INDUSTRIES LTD

(Jul-13-Nov-14)

Hands on worked on various data initiatives to optimize tire manufacturing processes

Industrial Analytics

- ✓ Data scientist, responsible for controlling and optimisation of radial building process using machine learning algorithms
 - o Involved **Tire Uniformity Data Analysis**, High level analysis on tyre technology through **data pattern study**, patents and company database
 - o Tools Used: **Power BI, SAS and Excel**

EDUCATION SUMMARY

EXAMINATION	SCHOOL/COLLEGE	UNIVERSITY/BOARD	%/CGPA	YEAR
PGD-IRI	Indian Rubber Institute	Indian Institute of Technology Kharagpur	CGPA 6.82	2014
B.E (PSCT)	Delhi College of Engineering	Delhi University	76%	2013
XII	Jaspal Kaur Public School	C.B.S.E	78.40%	2008
X	Jaspal Kaur Public School	C.B.S.E	76.40%	2006

Certifications:

1. AWS Certified Solutions Architect – Associate - AWS
2. Certification Programme in Big Data Engineering with Hadoop and Spark - IIT Roorkee
3. Certification Programme in Python for Machine Learning - IIT Roorkee
4. Certification Programme in Machine Learning Specialization - IIT Roorkee
5. Certification Programme in Deep Learning - IIT Roorkee